

AMUTHEEZAN SIVAGNANAM

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EDUCATION

PhD in Informatics Pennsylvania State University, University Park, PA Supervisor: Dr. Aron Laszka	Aug. 2022 - Present
PhD. in Computer Science (Transferred to PSU) The University of Houston, Houston, TX Supervisor: Dr. Aron Laszka	Aug. 2019 - Aug. 2022
MSc. in Computer Science University of Houston, Houston, TX, GPA : 4.00 /4.00	Aug. 2019 - Aug. 2022
BSc. (Hons) in Computer Science and Engineering Faculty of Engineering, University of Moratuwa, Sri Lanka, GPA : 3.81 /4.20	Jan. 2014 - Jan. 2018

SKILLS

Fields : Artificial Intelligence, Machine Learning, Reinforcement Learning, Combinatorial Optimization, Operational Research
Language and Tools : Python, TensorFlow, PyTorch, Keras, Git, Docker, CPLEX, Mosek, Google OR-Tools, C, C++, Java

SELECTED PUBLICATIONS

- **Amutheezan Sivagnanam**, Ava Pettet, Hunter Lee, Ayan Mukhopadhyay, Abhishek Dubey, Aron Laszka. “*Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing*” Accepted to be present in the Proceedings of the 41st International Conference on Machine Learning, Vienna, Austria. PMLR 235, 2024 (**ICML 2024**). [Link](#)
- **Amutheezan Sivagnanam**, Salah Uddin Kadir, Ayan Mukhopadhyay, Philip Pugliese, Abhishek Dubey, Samitha Samaranyake, Aron Laszka. “*Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit*” In Proceedings of the 31st International Joint Conference on Artificial Intelligence (**IJCAI 2022**). [Link](#)
- **Amutheezan Sivagnanam**, Afiya Ayman, Michael Wilbur, Philip Pugliese, Abhishek Dubey, Aron Laszka. “*Minimizing Energy Use of Mixed-fleet Public Transit for Fixed-route Service.*” In Proceedings of the 35th AAAI Conference on Artificial Intelligence (**AAAI 2021**), February 2–9, 2021, Virtual Event, 7 pages. [Link](#)
- Soodeh Atefi, **Amutheezan Sivagnanam**, Afiya Ayman, Jens Grossklags, Aron Laszka. “*The Benefits of Vulnerability Discovery and Bug Bounty Programs: Case Studies of Chromium and Firefox*”. Published in Proceeding of The Web Conference 2023 (**WWW 2023**), April 30 - May 4, 2023, Austin, TX, USA, 8 Pages. [Link](#)

RESEARCH EXPERIENCE

Graduate Research Assistant — Pennsylvania State University Applied Artificial Intelligence Lab	University Park, PA, USA Aug. 2022 - present
• Proposed Deep-Reinforcement Learning based solution for transportation problem such as Stationing and Routing and implemented the approaches in python using libraries such as Google OR Tools, and TensorFlow.	
Graduate Research Assistant — The University of Houston Resilient Networks and Systems Lab	Houston, TX, USA Sep. 2019 - Aug. 2022
• Introduced mathematical models to solve transit optimization problems (e.g., minimizing energy consumption in public transportation operating mixed fleets of electric and gasoline vehicles, optimizing the online booking for paratransit services with offline vehicle routing problem setting).	
• Proposed Deep-Reinforcement Learning based solution for transportation problem such as Scheduling and Booking and implemented the approaches in python using libraries such as CPLEX, Google OR Tools, and TensorFlow.	
• Collected the real-world data and analyzed it to study trends with the assistance of Python (e.g., para-transit operation before and after COVID-19, the benefit of vulnerability reward programs)	

SELECTED PHD PROJECTS

Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service

- Introduce a novel mathematical model to minimize the energy consumption of public transit agencies that operate mixed fleets of Electric (EV) and Internal Combustion Engine (ICEVs) Vehicles.
- Devise a novel solution approach based on mixed-integer-programming to obtain optimal solution and introduce polynomial-time heuristics/meta-heuristics to solve larger problem instances

Offline Vehicle Routing Problem with Online Bookings for Paratransit Operations

- Introduce a novel mathematical formulation that handles both large number of requests and real-time decision.
- Propose a novel solution approach using deep-reinforcement learning support by anytime algorithm for real-time decision-making that contributes towards minimizing the operational costs

Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing

- Propose a novel deep-reinforcement learning-based solution approach replacing the online-search in current state-of-the-art to make reposition decision in fraction of seconds
- Using transformers to handle the variable-dimensional state and action space

Online Vehicle Routing Problem with Advance Booking

- Propose a novel solution approach that uses exhaustive search insertion and anytime algorithm to solve the problem of advance booking